

RUNTIME ENVIRONMENT

• Runtime environment :

• Storage allocation strategies =

1) Static :

- ① allocation is done at compile time.
- ② Bindings do not change at runtime.
- ③ one activation record per procedure.

Dis :

- ① Recursion is not supported.
- ② size of data objects must be known at compile time.
- ③ Data structure can't be created dynamically.

2) Stack :

- ① Whenever a new activation begins, activation record is pushed on to the stack and whenever activation ~~record ends~~ ends, activation record is popped off.

- ② Local variables are bound on fresh storage.

Dis :

- ① Local variable cannot be ~~ret~~ retained once activation ends.

3) Heaps :

- ① Allocation and deallocation can be any order.

Dis : ① Heap management is overhead.

• Summary :
activation can have -

- ① permanent lifetime in case of static allocation.
- ② Nested lifetime in case of ~~static~~ stack allocation.
- ③ Arbitrary life time in case of heap allocation.

